RAJA KUMAR

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Summary

Final-year B.Tech CSE student (CGPA 9.11) specializing in AI/ML, Generative AI, and intelligent systems. Experienced in building RAG applications, AI agents & vector databases, using Python, FastAPI, Docker, and CI/CD.

Skills

Programming Languages: Java, Python, C

Backend & Frameworks: FastAPI, Flask, Express, REST APIs, SDKs, Backend Development, Next.js, React

AI/ML Generative AI: Machine Learning, Deep Learning, Generative AI, LLMs, AI Agents, RAG, Prompt Engineering, LangChain, LangGraph, TensorFlow, Evaluation Methods

Cloud, DevOps Observability: AWS, Docker, Git, GitHub Actions, CI/CD, Telemetry, Observability

Software Engineering: System Design, Software Design, Data Structures Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Debugging

Responsible AI Security: Responsible AI, AI Security, Reliability

Projects

Yseek: YouTube Video RAG Engine | Python, Qdrant, faster-whisper, Sentence-Transformers Github

- Processed **200+ hours** of video via **faster-whisper** and **yt-dlp**, engineering a 75s time-window chunker to preserve temporal contexts.
- Built a **Qdrant** retrieval engine indexing **30,000+ vectors**, delivering **250ms** latency and **91% Hit@3** accuracy on multilingual queries.
- Integrated **FastAPI** and the **YouTube IFrame API** for instant seek-to-second playback and grounded LLM synthesis.

OmniAgent: Multi-Agent Trading System | Python, LangGraph, LangChain, Flask Github

- Architected a multi-agent AI trading platform utilizing LangGraph, automating **100%** of routine market analysis and trade orchestrations.
- Integrated GPT-4o, LLaVA, and TA-Lib indicators (RSI, MACD), reducing false trading signals by 30%.
- Constructed high-throughput Flask REST APIs to handle **50+** concurrent agent requests with **<200ms** inference latency.
- Automated cloud infrastructure deployment via Docker and GitHub Actions, cutting release cycles from days to **under 15 minutes**.

Rag-Db: RAG Visualizer & VectorDB Engine | Python, HNSW, Ollama, Docker, GitHub Actions Github

- Engineered a custom Vector Database in Python using HNSW ($O(\log N)$) and KD-Trees, processing **10,000+** embeddings with **sub-50ms** query latency.
- Deployed an air-gapped RAG pipeline, improving context retrieval accuracy by **40%** for AI-powered question answering.
- Developed a PCA-based visualization dashboard to benchmark **3** distinct search algorithms, reducing performance profiling time by **2 hours** per cycle.
- Streamlined application deployment with Docker Compose, accelerating CI/CD pipeline builds by **25%** via GitHub Actions.

Education

KIIT UNIVERSITY 2023 – 2027

B.Tech in Computer Science and Engineering **CGPA: 9.11**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Discrete Mathematics, Machine Learning, Data Mining

Work Experience

InAmigos Foundation | *Machine Learning Intern* Aug 2026 – Present

- Developed end-to-end ML pipelines covering **data preprocessing, EDA, missing-value handling, feature engineering, encoding, and feature scaling** using Python and Scikit-learn.
- Built and benchmarked **Linear Regression, Decision Tree, and Random Forest** models using **cross-validation, RMSE, MAE, R^2, GridSearchCV, and hyperparameter tuning**.

Achievements & Certifications

- Earned **3** professional credentials including AWS Cloud Architecting, AICTE Data Science Internship (Siemens), and Data Science, AI/ML & GenAI, covering **100+ hours** of advanced curriculum.
- Solved 500+ DSA problems and secured 1st Runner-Up at KIIT Hackathon UiPath Resume Screener.